

Project Design Phase

Phase 3 of 8 - Project Documentation

Project: OptiCrop: Smart Agricultural Production Optimization Engine

System Architecture

- Dataset (Crop_recommendation.csv) - 2,200 records across 22 crop types, each described by N, P, K, temperature, humidity, pH, and rainfall.
- Training Notebook - performs EDA (univariate, bivariate, multivariate analysis), outlier handling, seasonal crop extraction, K-Means clustering, and Logistic Regression training.
- Serialized Model (model.pkl) - the trained Logistic Regression classifier.
- Flask Application - loads the model and exposes Home, About, and Find Your Crop routes.
- User Interface - three HTML templates styled to match the OptiCrop visual identity (green agricultural theme).

Entity-Relationship Diagram

The diagram below models the full relational design for a multi-user version of OptiCrop, covering Users, SoilData submissions, Crop records, Datasets, ML Models, Predictions, and generated Reports - a natural extension path beyond the current single-model Flask deployment.

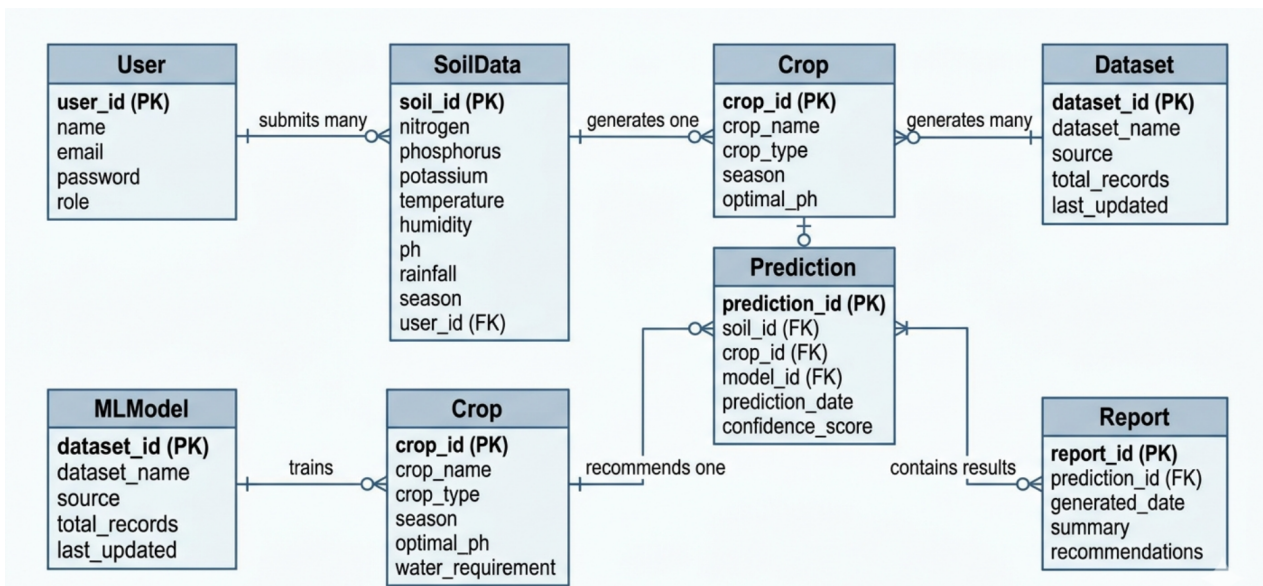


Figure 1: Entity-Relationship Diagram for the OptiCrop platform.

User Flow

- 1. User lands on the Home page and sees the OptiCrop introduction and navigation.
- 2. User optionally visits the About page to read the full project description.
- 3. User navigates to Find Your Crop and enters Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall values.
- 4. User clicks Predict; the Flask backend converts the inputs to a feature array and calls the trained model.
- 5. The recommended crop is displayed directly on the Find Your Crop page.